THE PROJECT TITLE

ON

Network Intrusion Detection Using Random Forest and Deep Neural Networks

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF __CSE_____

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1. Abstract:

The Network Intrusion Detection Using Random Forest and Deep Learning project aims to develop a machine-learning-based system for detecting malicious network traffic. The system preprocesses network traffic data, extracts relevant features, and classifies the traffic as normal or intrusive. Random Forest is used because it provides reliable classification and can handle a large number of features. A Deep Neural Network (DNN) is also used to learn complex patterns from network traffic. The performance of both models can be compared using accuracy, precision, recall, F1-score, and confusion matrix. The proposed system can help improve network security by providing an automated approach for identifying suspicious network activities.

2. INTRODUCTION:

Network security is an important part of modern computer systems because networks are continuously exposed to threats such as denial-of-service attacks, unauthorized access, scanning, malware, and other malicious activities. A Network Intrusion Detection System (NIDS) monitors network traffic and identifies whether the traffic is normal or potentially malicious.

Traditional intrusion detection techniques may have difficulty detecting complex and evolving attacks. Machine learning and deep learning can analyze large amounts of network traffic and identify patterns associated with attacks. This project uses Random Forest and Deep Learning techniques to build an intelligent intrusion detection system. The models can be trained using network-security datasets and evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

3. SOFTWARE REQUIREMENTS:

The software requirements for the Network Intrusion Detection Using Random Forest and Deep Learning project include Windows or Linux operating system and Python 3.x as the primary programming language. Jupyter Notebook, Google Colab, VS Code, or PyCharm can be used for developing and executing the project. Python libraries such as NumPy and Pandas are required for data processing and analysis, while Scikit-learn is used to implement the Random Forest algorithm and evaluate the machine learning model. TensorFlow/Keras is used to develop and train the Deep Neural Network model. Matplotlib and Seaborn can be used to visualize the dataset, model performance, and confusion matrix. Datasets such as NSL-KDD, UNSW-NB15, or CICIDS2017 are required for training and testing the intrusion detection models.

Signature of the Guide Course Instructor